PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP & Document AI

PROFESSIONAL SUMMARY

Machine Learning Engineer with experience developing NLP and document AI systems for Croatian-language administrative and customer-support text. Work includes document classification serving approximately 15,000 documents per day and Croatian-language NER development, model serving and monitoring, and Azure-based deployment. Doctoral research in Croatian legal-text NER is in progress (expected 2027). LLM experience is limited to a personal weekend RAG prototype; no production LLM work or LLM fine-tuning.

CORE SKILLS

Python; NLP; document classification; named entity recognition (NER); PyTorch; Hugging Face Transformers; scikit-learn; FastAPI; model serving; model monitoring; MLflow; Azure ML; Azure App Service; Azure Functions; Docker; PostgreSQL; pandas; spaCy; Label Studio; Evidently; Oracle SQL; fastText; sentence-transformers; FAISS

PROFESSIONAL EXPERIENCE

ML Engineer | Slavonica Document AI *2023–Present*

- Develop and operate machine learning systems for document classification and named entity recognition in Croatian public-administration documents, including model serving and monitoring.
- Built an incoming-document classifier for 23 classes, achieving 94% accuracy and 0.91 macro F1; fine-tuned XLM-R base and served the model as a FastAPI service in Docker on Azure App Service, processing approximately 15,000 documents per day, using PyTorch, Hugging Face Transformers, Azure ML and MLflow.
- Developed a Croatian NER model for persons, organisations, OIB numbers, addresses and legal references by fine-tuning BERTić.
- Created an annotation guideline and dataset of 4,000 documents with two annotators; achieved inter-annotator kappa of 0.87.
- Implemented weekly model-monitoring reports using Evidently for input text length, vocabulary and predicted-class distribution, with PSI > 0.2 alerts. Detected a silent client OCR change that reduced accuracy by 9 percentage points; the issue was fixed within a week.

Data Scientist | Baranjska kreditna banka *2021–2023*

- Worked on customer churn prediction and complaint classification.
- Built a customer complaint topic classifier using a TF-IDF and logistic-regression baseline, followed by fastText, to classify 11 topics for weekly reporting to customer service management.

SELECTED PROJECT

Retrieval-Augmented Generation (RAG) Question Answering Demo | Personal project
2025

- Built a weekend prototype for question answering over Croatian laws using a sentence embedding model and an open-weight LLM accessed through an API.
- The demo was not used in production and was checked manually; no LLM fine-tuning or production LLM experience.
- Tools: Python, sentence-transformers, FAISS.

EDUCATION

Doctoral Studies, University Drava *In progress; expected 2027*

Research topic: Named entity recognition in Croatian legal texts

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate *2024*

ADDITIONAL CONTRIBUTIONS

- Guest lecture at an NLP summer school | 2025 (90 minutes)
- Published work on Croatian named entity recognition at a regional workshop | 2024
- Uses English in regular meetings with a client from Slovenia